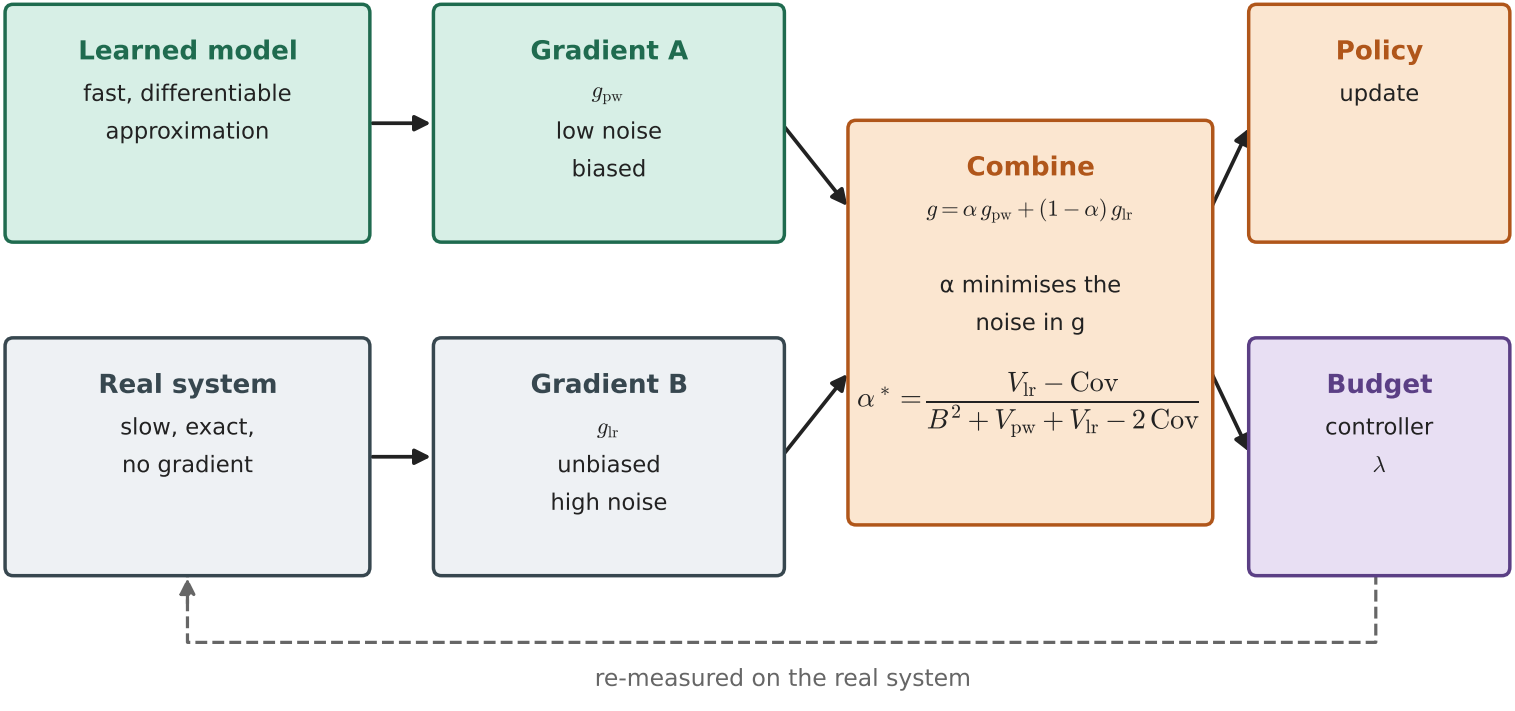
Learned model
fast, differentiable
approximation
Gradient A
g_{pw}
low noise
biased
Policy
update
Combine
$g = \alpha\, g_{\mathrm{pw}} + (1-\alpha)\, g_{\mathrm{lr}}$
α minimises the
noise in g
$\alpha^* = \frac{V_{\mathrm{lr}} - \mathrm{Cov}}{B^2 + V_{\mathrm{pw}} + V_{\mathrm{lr}} - 2\,\mathrm{Cov}}$
Real system
slow, exact,
no gradient
Gradient B
g_{lr}
unbiased
high noise
Budget
controller
λ
re-measured on the real system